7-feb-2020

FUNCTIONS

Basics.

(1) Square root

$\sqrt{\ }$ is positive quantity

$\sqrt{16} = 4$
$\neq -4$

$x^2 = 16$ (Taking square root on both sides

$x = \pm 4$
↳ This $\pm$ is bcz of square root on both side.

$\sqrt{16} \neq -4$.

(2) $\sqrt{f(x)}$ exist only when $f(x) \geq 0$

$\sqrt{x^2-16}$

$x^2 - 16 \geq 0$

$(x-4)(x+4) \geq 0$

+   −   +
-4 $+4$

$(-\infty, -4] \cup [4, \infty)$

(3) $x^2, x^4, x^6 \ldots\ldots x^{2n}$ is Non negative (0 & positive)

if $x \neq 0$, then +ve

(4) $2^x, 3^x, 4^x \ldots$ Exponential function

$2^x > 0$ always +ve

(5) $|x| \longrightarrow$ Non negative

$|x| \geq 0$

(6) $\sqrt{u} + |v| + w^2 = 0$

$u = 0$
$v = 0$
$w = 0$

If sum of 2 or more positive quantity is equal to zero, then they all are equal to zero

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product of roots ↓ Sum of roots ↓

(14) (i) $x^3 - 6x^2 + 11x - 6 > 0$

↳ Linear factor

$(x-1)(x-2)(x-3) > 0$

Yaad hone chahiye

$-$ $+$ $-$ $+$

1   2   3

$(1,2) \cup (3,\infty)$

Power.
Even → Same sign
Odd → change sign

(ii) $(x-1)^2 (x+4)^1 < 0$

1, 4

$-$ $+$ $+$

-4 1

$(-\infty, -4)$

(iii) $(x-1)^2 (x-2)^3 (x-3)^5 \geq 0$

→ or agar equal to ($\geq$) 0 hota. toh closed interval

$+$ $+$ $-$ $+$

1   2   3

$(-\infty,1) \cup (1,2) \cup (3,\infty)$

(iv) $\dfrac{(x-1)^2 (x-3)^3}{(x-5)^5} \geq 0$

$(x-5)^5$ → open interval

$+$ $+$ $-$ $+$

1   3   5

$(-\infty, 3] \cup (5, \infty)$

(v) $\sqrt{\dfrac{(x-1)^2 (x-3)^3}{(x-5)^5}} \geq 0$

$\sqrt{f(x)}$ $\forall\ f(x) \geq 0$

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(11) If $A \in [1,4]$, $A^2 \in [1,16]$

$A \in [-1,4]$ $A^2 \in [0,16]$

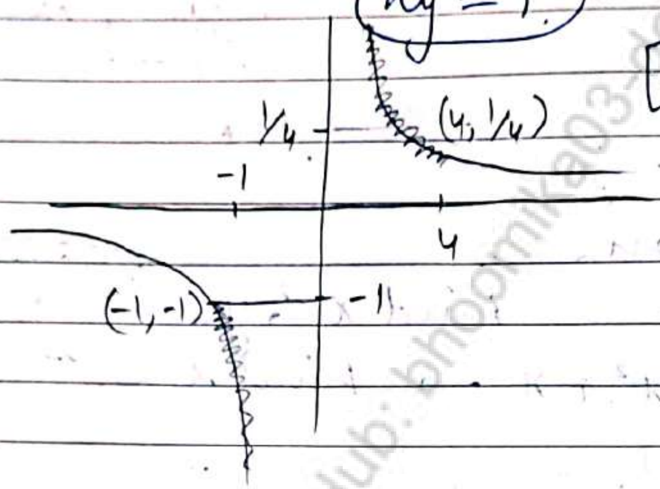
$A \in [-1,4]$ $\frac{1}{A} \in [\cancel{(-1,0) \cup (0,1/4)}]$

$y = x$

$y = 1/x$ rectangular hyperbola.

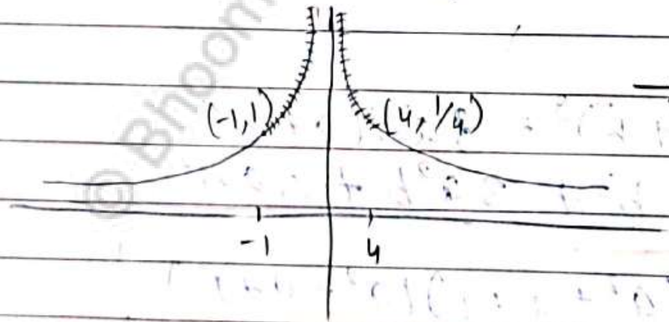
$xy = 1$

$[-\infty, -1] \cup [1/4, \infty)$



(12) $y = \frac{1}{|x|}$ ~~$x \in [-1;4]$~~ $(-1,0) \cup (0,4]$

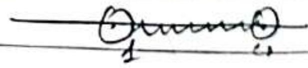
$y \in ?$ $[\frac{1}{4}, \infty)$



$\rightarrow [1, \infty)$

$[1/4, \infty)$

(13) Open - Interval $\rightarrow x \in (1,4)$



Close - Interval $\rightarrow x \in [1,4]$

$x \geq 1 \rightarrow$

$\rightarrow \{1, 2, 3\}$

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product of roots   Sum of roots

(14) (i) $x^3 - 6x^2 + 11x - 6 > 0$

Linear factor

$(x-1)(x-2)(x-3) > 0$ yaad hone chahiye

$-\quad 1 \quad + \quad 2 \quad - \quad 3 \quad +$

$(1,2) \cup (3,\infty)$

Power

Even → Same sign
Odd → change sign

(ii) $(x-1)^2 (x+4)^1 < 0$

$1, 4$

$-\quad -4 \quad + \quad 1 \quad +$

$(-\infty, -4)$

(iii) $(x-1)^2 (x-2)^3 (x-3)^5 \geq 0$

Or agar equal to ($\geq$)0 hota to closed interval

$+\quad 1 \quad + \quad 2 \quad - \quad 3 \quad +$

$(-\infty, 1) \cup (1,2) \cup (3,\infty)$

(iv) $\dfrac{(x-1)^2 (x-3)^3}{(x-5)^5} \geq 0$

$(x-5)^5$ → open interval.

$+\quad 1 \quad + \quad 3 \quad - \quad 5 \quad +$

$(-\infty, 3] \cup (5, \infty)$

(v) $\sqrt{\dfrac{(x-1)^2 (x-3)^3}{(x-5)^5}} \geq 0$ $\sqrt{f(x)}$ $\forall f(x) \geq 0$

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RULES

*1 Factorize the given equation & find all odd powers
*2 get value from bracket & arrange them on number line
*3 Find value of x for which function is not defined, & remove them.
* (Values / Roots of denominator or even power factor of Numerator)
*4 Put the sign to right most side & then change of sign for odd powers bracket only.
*5 For $>, \geq$ opt (+ve) sign
$<, \leq$ opt (-ve) sign

Coeff of x in all factors must be 1

Q) $(-x^2+16 \geq 0)$

Sol $[-4, 4]$

$16-x^2 \geq 0$

$(4-x)(4+x) \geq 0$

$-(x-4)(x+4) \geq 0$

$(x-4)(x+4) \leq 0$

$+ \quad -4 \quad - \quad 4 \quad +$

Q) $(2x-3)(4x-16) \leq 0$

$x = 3/2, \quad x = 4$

$+ \quad 3/2 \quad - \quad 4 \quad +$

$[3/2, 4]$

correct way

$2(x-3/2)4(x-4) \leq 0$

$8(x-3/2)(x-4) \leq 0$

Coeff of $x \rightarrow 1$ bna do

$$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ac$$
$$= \Sigma a^2 + 2\Sigma ab$$

$$a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$
$$= (a+b+c)\left[\frac{(a-b)^2+(b-c)^2+(c-a)^2}{2}\right]$$

$$(a-b)^2+(b-c)^2+(c-a)^2 \geq 0$$
$$a^2+b^2+c^2-ab-bc-ca \geq 0$$

$$a^3+b^3+c^3 = 3abc \quad \boxed{\begin{array}{l}\text{If } a+b+c=0 \\ \text{or} \\ a=b=c\end{array}}$$

Note: $(\ )_1^3 + (\ )_2^3 + (\ )_3^3 = 3(\ )_1(\ )_2(\ )_3$

$$\left[\begin{array}{l}(\ )_1+(\ )_2+(\ )_3 = 0 \\ (\ )_1 = (\ )_2 = (\ )_3\end{array}\right]$$

$\to 3(\ )^3$ ho jayega.

If $\sin^4 x + \cos^6 x = 1$ Find x if $x \in [0, \pi/2]$

Sol

$\sin^2 x > \sin^4 x$

$\cos^2 x > \cos^6 x$

$\boxed{0 < x < 1}$

Toh jitni power badhaoge utni choti value ayegi

$\star$ $x^n > x^{2n}$

$\sin^4 x + \cos^6 x \leq \sin^2 x + \cos^2 x$

$\boxed{\sin^4 x + \cos^6 x \leq 1}$

$\sin^2 x = \sin^4 x$ $\qquad$ $\cos^2 x = \cos^4 x$

$(0, 1)$ $\qquad$ $(0, 1)$

$\sin x = 0, 1$ $\quad [x = 0, \pi/2]$ $\qquad$ $\cos x = 0, 1$ $\quad [x = 0, \pi/2]$

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PRIME NUMBERS

Smallest prime no → 2
Only even prime no → 2
Prime → odd.

PERFECT SQUARE

$① = 4K$ or $4K+1$ $\quad \cancel{4K+2} \quad \cancel{4K+3}$

$= 3n$ or $3n+1$

NUMBER SYSTEM

$\mathbb{N} \rightarrow \{1, 2, 3 \ldots \infty\}$
$W \rightarrow \{0, 1, 2 \ldots \infty\}$
I or $Z \rightarrow \{-\infty \ldots, -1, 0, 1, \ldots \infty\}$
$Q \rightarrow 0.$ ——— → Repeating → 7.6666
or
terminating → 6.341341

Q^c → Non terminating → Non Repeating.
π or e
$\pi \rightarrow 3.14$, $e \rightarrow 2.718$

$\pi \rightarrow 3.14$
$\pi^2 \rightarrow 9.86$ $\qquad \frac{\pi}{2} \rightarrow 1.57.$
$\frac{\pi}{4} < 1$

CO PRIME NUMBERS

H.C.F $(a,b) = 1$, then it is coprime No.
$(4,5) = 1 \quad (6,7) = 1 \quad (8,9) = 1$

* Two consecutive no. is always co-prime

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TWIN - PRIME,
Difference of two prime no = 2
(3,5) (5,7) (11,13) (17,19)

LCM of (π, π^2)

Q. $\frac{2x-3}{3x-5} \geq 3$. Solve for x.

$(x \neq 5/3)\ (x \neq 3/2)$

sol

$\frac{2x-3}{3x-5} - 3 \geq 0$

$\frac{2x-3-3(3x-5)}{3x-5} \geq 0$

$\frac{2x-3-9x+15}{3x-5} \geq 0$

$\frac{-7x+12}{3x-5} \geq 0$

$\frac{-7(x-12/7)}{3(x-5/3)} \geq 0$

$\frac{x-12/7}{x-5/3} \leq 0$

$1.7 \rightarrow 12/7$
$1.6 \rightarrow 5/3$

so

$+$ $5/3$ $-$ $12/7$ $+$

$\left(\frac{5}{3}, \frac{12}{7}\right]$

LCM & HCF

1. HCF $(2,4,6) = 2$
2. $(2,4,5) = 1$
3. LCM $(2,4,8) = 8$

   $(3,6,9) = 18$

   $LCM(2, 2^2, 2^3 \ldots 2^n) = 2^n$

   $LCM(3, 3^2, 3^3 \ldots 3^n) = 3^n$

   $LCM(n!, (n+1)!) = (n+1)!$

4. ~~$LCM(\pi, \pi^2)$ = Not possible.~~

   $LCM(\pi, 2\pi) = 2\pi$

Result: $L.C.M(a,b) = c$, then $\frac{c}{a}$, $\frac{c}{b}$ should be integer.

5. LCM $(\sqrt{3}, \sqrt{6})$ = Not possible
6. $LCM(\sqrt{3}, 4\sqrt{3}) = 4\sqrt{3}$.

   $(\alpha, K\alpha) = K\alpha$

   K → scalar integer

☆ $LCM\left\{\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right\} = \frac{L.CM\{a,c,e\}}{H.CF\{b,d,f\}}$

ex $\left\{\frac{\pi}{3}, \frac{2\pi}{4}, \frac{3\pi}{5}\right\} = \underline{3\pi}$

$\left\{\frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{5}\right\}$

LCM $(-4, 4) \rightarrow 4$.

Absolute value ka niklta hai

RATIO AND PROPORTION

Componendo & Dividendo

*1 If $\frac{a}{b} = \frac{c}{d}$

then $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

// $\frac{\text{Numerator} + \text{denominator}}{\text{Numerator} - \text{denominator}}$

*2 $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{\alpha a + \beta c + \gamma e}{\alpha b + \beta d + \gamma f}$.

Q Find Numerical value of ratio

$\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = ?$

Sol. $\frac{a+b+c}{b+c+c+a+a+b} = \frac{a+b+c}{2a+2b+2c} = \frac{1}{2}$.

LOGARITHM

$\log_b a$ is defined. (a: argument, b: base)

$a > 0,\ b > 0,\ b \neq 1$

Property

① $\log_b a = K$ iff $\boxed{a = b^K}$

② $\log_a a = 1$ iff $\boxed{a = b}$

③ $a^{\log_a b} = b^{\log_a a} = b^1$

$= b$

agr yeh same hua toh bcha hua he ayega.

④ $\log_a (bc) = \log_a b + \log_a c$

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ki power age a sakti hai

(5) $K \log_a b = \log_a b^{(K)}$

(6) $\log_{b^m} a = \frac{1}{m} \log_b a$

base ki power hua toh denominator mein a jayega, argument ki power numerator mein

(7) $\log_{b^m} a^n = \frac{n}{m} \log_b a.$

(8) $\log_a(b) - \log_a(c) = \log_a(b/c)$

(9) Base Theorem

$$\log_b a = \frac{\log_c a}{\log_c b}$$

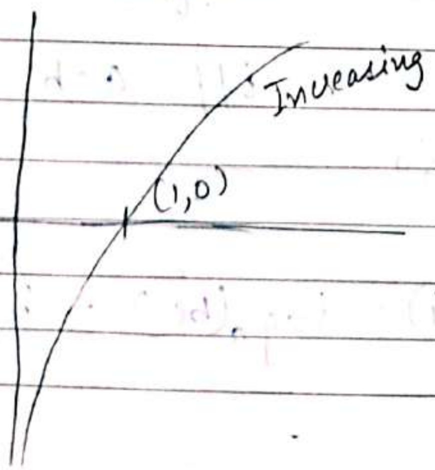
(10) $$\log_b a = \frac{1}{\log_a b}$$

$$\log_{1/a} b = \log_{a^{-1}} b = -\log_a b$$

Any positive Number 'K' can be written as ~~$K = \log_a K$~~

$$K = a^{\log_a K}$$

GRAPH



$y = \log_b x$

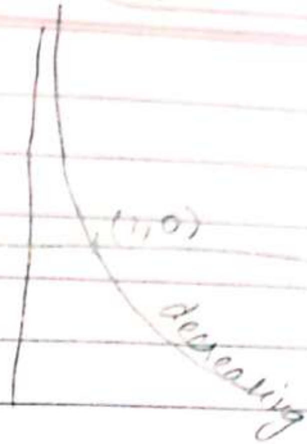
$\boxed{x = b^y}$

$b > 1$

$0 < b < 1$

$y = \log_b x$

$x = b^y$



LAWS OF EXPONENT

* Product Rule

$$a^x \cdot a^y = a^{x+y}$$

* Quotient Rule

$$\frac{a^x}{a^y} = a^{x-y}$$

* Power of Power

$$(a^x)^y = a^{xy}$$

* Zero exponent

$$a^0 = 1$$

* Negative exponent

$$a^{-x} = \frac{1}{a^x}$$

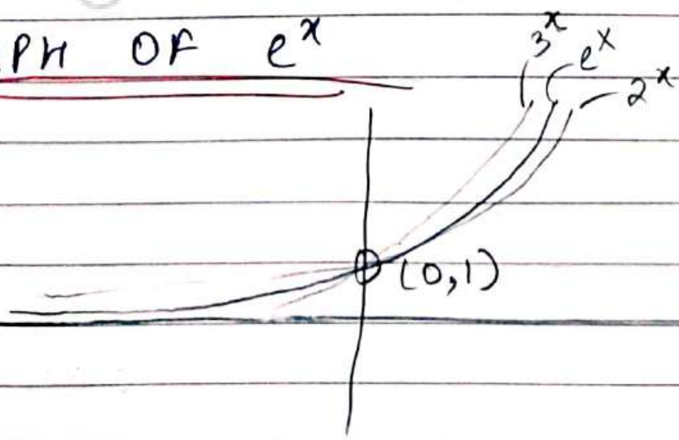
* Power of product

$$(ab)^x = a^x b^x$$

* Power of quotient

$$\left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}$$

GRAPH OF e^x



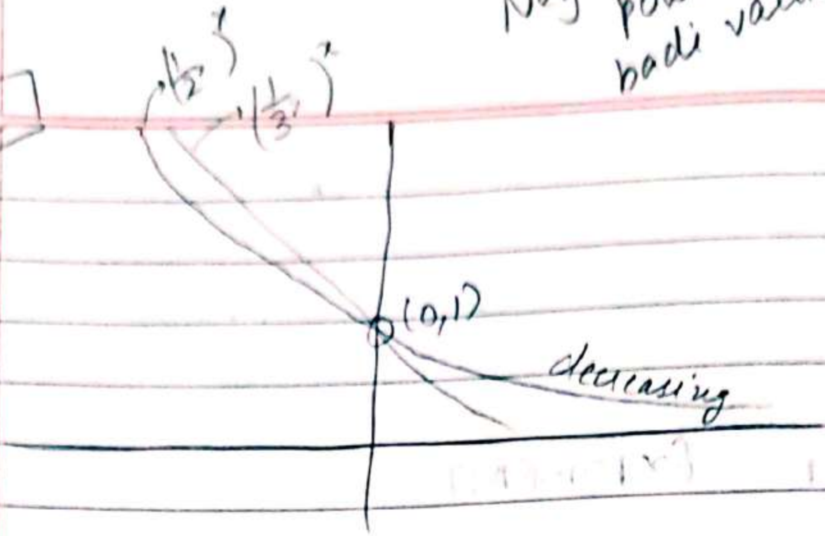
$\boxed{y = e^x}$

Increasing if base $a > 1$

Negative power for badi value

$\boxed{0<a<1}$

$y = \left(\frac{1}{2}\right)^x$



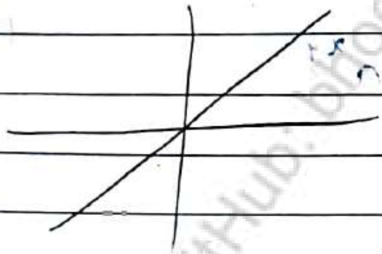
$\left(\frac{1}{3}\right)^{-2} > \left(\frac{1}{2}\right)^{-2}$

MODULUS

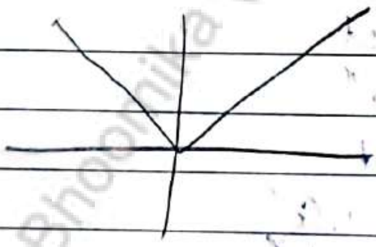
(i) $|x| = 2$

$x = \pm 2$

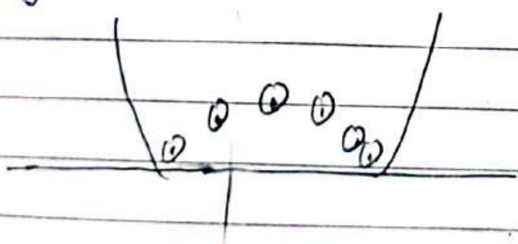
(ii) $y = x$



(iii) $y = |x|$



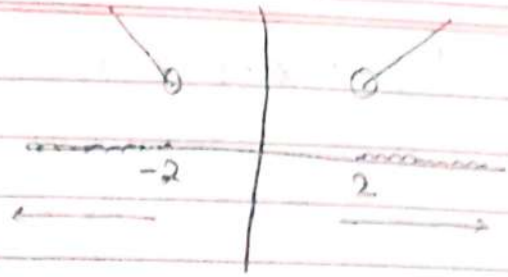
(iv) $y = |ax^2 + bx + c|$ $\quad a>0 \quad D>0$ → 2 points pe cut Krega



(10) $|x| > 2$

$y = |x| > 2$

$y > 2$



$|x| > 2$

$x \in (-\infty, -2) \cup (2, \infty)$

$|x| \leq a \rightarrow -a \leq x \leq a$

$|x| \geq a \rightarrow x \geq a$ or $x \leq -a$.

Q. $|x| > -2$.

$x \in \mathbb{R}$

Q. $|x| < -5$

$x \in \phi$ No real value of x.

Q. $a < |x| < b$

$a < |x|$ & $|x| < b$

$\boxed{x \geq a \text{ or } x \leq -a}$ $\boxed{-a < x < a}$

$+2 < |x| < 3$

$2 < |x|$ & $|x| < 3$

$(-3, -2) \cup (2, 3)$

$\underbrace{-2 < |x|}_{\text{Ignore}} < 3 \rightarrow (-3, 3)$

$|xy| = |x||y|$

$y = |x|$

~~$|x| = |y|$~~

$\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$

$y = |x| = \begin{cases} x & x \geq 0 \\ -(-x) & x < 0 \end{cases}$

$|x|^2 = x^2$

$|y| = |x|$

$|x| = y$ $\quad$ $|y| = x$

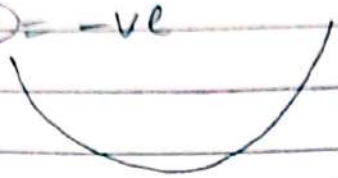
$x = \pm y$ or $y = \pm x$

$|x^2+x+1|$

$|f(x)| = \begin{cases} f(x) & f(x)\geq \\ -f(x) & f(x)<0 \end{cases}$

$a=1>0$

$D=-ve$



Q $\left|\dfrac{x^2+1}{2x}\right|$

$\dfrac{|x^2+1|}{|2x|} = \dfrac{x^2+1}{2|x|} =$

Q $||||x|||| = |x|$

Q $-3 \leq x^2+x+2 \leq 3$

$|x^2+x+2| \leq 3$

Hint.

$\boxed{|x|<a \qquad -a<x<a}$

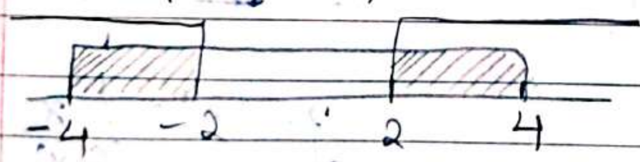
Q $|x-1| \leq 2$

$-2 \leq y \leq 2$

$-2 \leq x-1 \leq 2$

$-1 \leq x \leq 3$

Q $2<|x|<4$



Q $2 \leq |x-1| \leq 3$

$\quad 1$

Step 1. Phele zero find karo then add kedo

Step 2. fir backward chale jao.

$-3 \le x-1 \le 3$ & $x-1 \ge 2$ (or) $x-1 \le -2$

$-2 \le x \le 4$ & $x \ge 3$ (or) $x \le -1$

$[-2,-1] \cup [3,4]$

(1) $-1 < \dfrac{x^2}{3} \le 1$

$\underbrace{-1 < \dfrac{x^2}{3}}_{\text{Ignore.}}$

$\dfrac{x^2}{3} \le 1$

$x^2 \le 3$

$x \le \sqrt{3}$

$-\sqrt{3} \le x \le \sqrt{3}$

$-1 \le \sin\theta \le 1$

Domain of
(1) $\sin^{-1}(f(x))$
(2) $\cos^{-1}(f(x))$

$-1 \le f(x) \le 1$ & solve

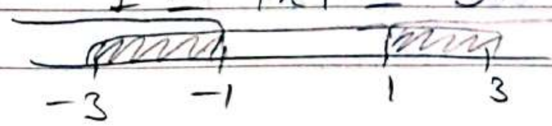
Domain : (i) $\sin^{-1}(x^2/3)$ $\quad -1 \le \dfrac{x^2}{3} \le 1$

$x \in [-\sqrt{3}, \sqrt{3}]$

(ii) $y = \cos^{-1}(|x|-2)$.

$-1 \le |x|-2 \le 1$

$1 \le |x| \le 3$



$(-3,-1) \cup (1,3)$

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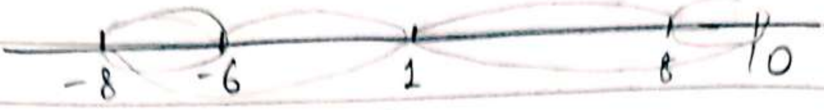
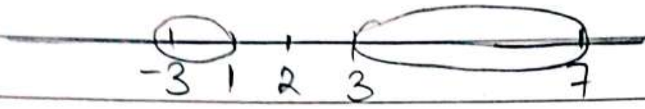
$7 \le |x-1| \le 9$.



$[-8,-6] \cup [8,10]$

$1 \le |x-2| \le 5$



$[-3,1] \cup [3,7]$

$\underline{-1 \le |x-2|} \le 5$

Ignore

$|x-2| \le 5$

$-5 \le y \le 5$

$-5 \le x-2 \le 5$

$-3 \le x \le 7$

$-x+2 \le 5$ $x-2 \le 5$

$-x \le 3$ $x \le 7$

$x \ge -3$

$[-3,7]$

Domain of $y = \sin^{-1}\left(\frac{|x|-2}{4}\right)$

$-1 \le \frac{|x|-2}{4} \le 1$

$-4 \le |x|-2 \le 4$

$\underline{-2 \le |x|} \le 6$

$|x| \le 6$

$-6 \le x \le 6$

$[-6,6]$

Domain of $y = \cos^{-1}(4x-1)$

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$$-1 \le 4x - 1 \le 1$$
$$0 \le 4x \le 2$$
$$0 \le x \le \frac{1}{2}$$
$$\left[0, \frac{1}{2}\right]$$

$y = \sqrt{\frac{2|x|-4}{6}}$ $\quad \sqrt{f(x)}$

$\cancel{y^2 = \frac{2|x|-4}{6}}$

$\cancel{y^2 = \frac{2\cdot|x|}{6} - \frac{4}{6}}$

$$f(x) \geq 0$$
$$\frac{2|x|-4}{6} \geq 0$$
$$2|x| - 4 \geq 0$$
$$2|x| \geq 4$$
$$|x| \geq 2$$
$$x \geq 2 \text{ or } x \le -2$$

Q $y = \sqrt{x^2 - x - 2}$

$\sqrt{f(x)}$

$f(x) \geq 0$

$$x^2 - x - 2 \geq 0$$
$$x^2 - 2x + x - 2 \geq 0$$
$$x(x-2) + 1(x-2) \geq 0$$
$$(x-2)(x+1) \geq 0$$

$[-\infty, -1] \cup [2, \infty)$

$+ \quad -1 \quad - \quad 2 \quad +$

Q Domain of $\sqrt{\frac{1-|x|}{2-|x|}}$

$\sqrt{f(x)} \qquad f(x) \geq 0$

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$$\frac{1-|x|}{2-|x|} \geq 0 \qquad \boxed{\text{Let } y=|x|}$$

$\frac{-(|x|-1)}{-(|x|-2)}$ ←

$$\frac{y-1}{y-2} \geq 0$$

+   −   +
1   2

$(-\infty, 1] \cup (2, \infty)$

$|x| > 2$ or $|x| \leq 1$

$(-\infty, -2) \cup (2, \infty) \quad \cup \quad [-1, 1]$

Q $y = |x-1| + |x-3|$ 2 constant   Domain.
$3-1$
$(-\infty, \infty) \in R$

$x-1=0$; $x-3=0$
$x=1$ $x=3$
yha pe −ve   yha pe add karna hai

jo zero kr raha hai wo plot kar lo.

$-\infty$ ← 1 — 3 → ∞

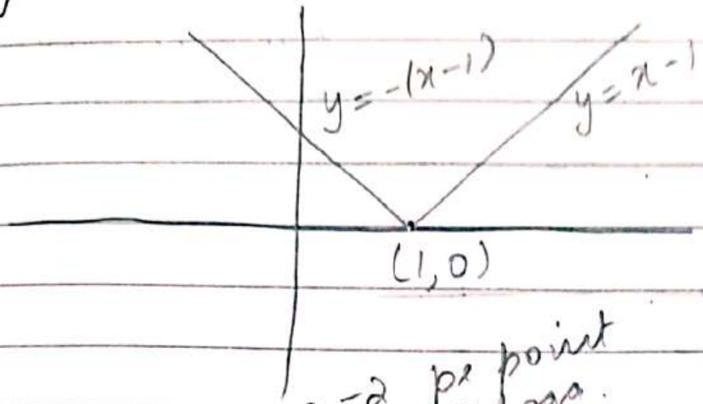
$$y = \begin{cases} -(x-1)-(x-3) & -\infty < x \leq 1 \\ (x-1)-(x-3) & 1 \leq x \leq 3 \\ (x-1)+(x-3) & 3 \leq x < \infty \end{cases}$$

$$y = \begin{cases} -2x+4, & -\infty < x \leq 1 \\ 2, & 1 \leq x \leq 3 \\ 2x-4, & 3 \leq x < \infty \end{cases}$$

The line $y=-2x+4$ cuts x-axis at $A(2,0)$ & y-axis at $B(0,4)$
$y=2$, this is a line parallel to x-axis
$y=2x-4$ cuts x-axis at $C(2,0)$ & y-axis at $D(0,-4)$

1 pe chale jayege wha se V banega.

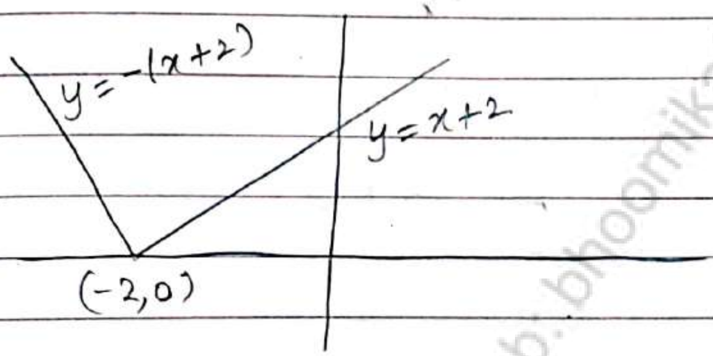
- $y = |x-1|$

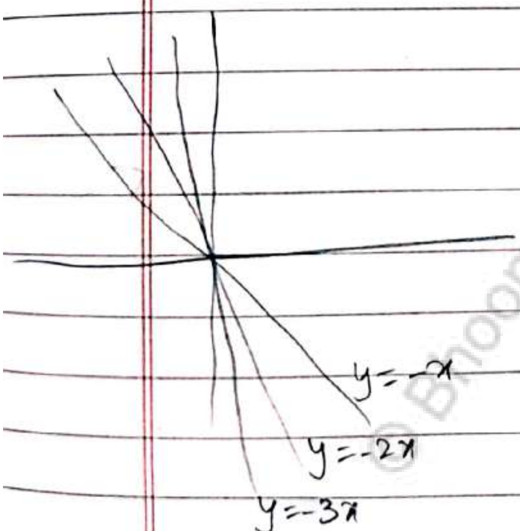


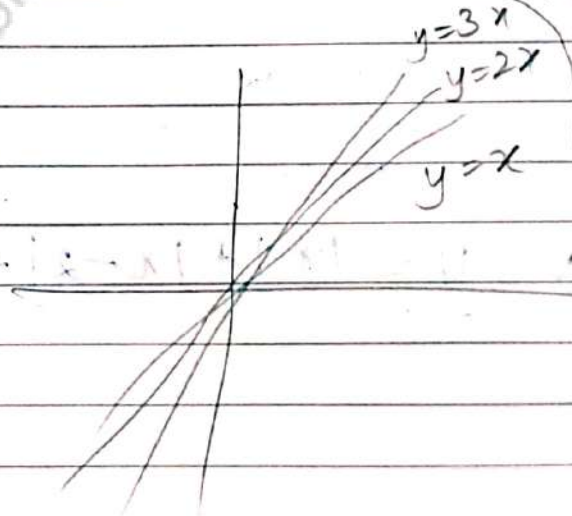
-2 pe point shift hoga.

- $y = |x+2|$

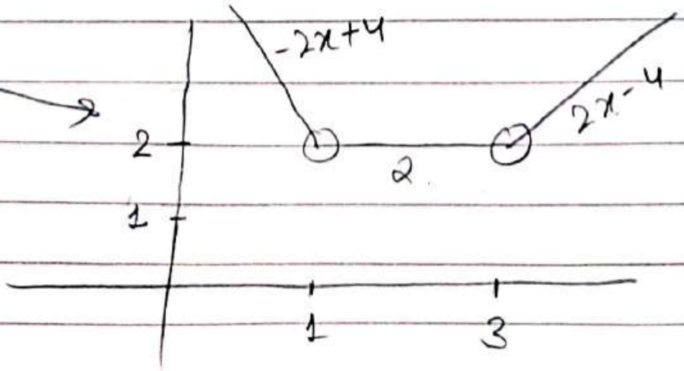
$$\begin{cases} x+2 & x \geq -2 \\ -(x+2) & x < -2 \end{cases}$$







Jese Jese slope increase krege x=0 pe bend krege



Domain $(-\infty, \infty)$

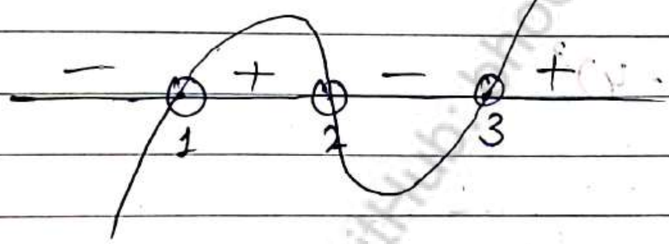
Range $[2, \infty)$

GRAPH CAN BE MADE BY FOLLOWING 3 METHODS

- Using Rules of Inequality (Wavy Curve Method)
- Using Transformation
- Using Calculus (AOD)

Q $y = x^3 - 6x^2 + 11x - 6$

$(x-1)(x-2)(x-3)$



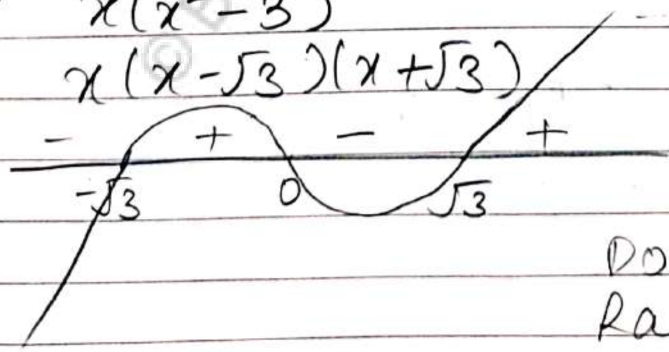
Domain $\rightarrow (-\infty, \infty)$

Range $\rightarrow (-\infty, \infty)$

Q. $y = x^3 - 3x$

$x(x^2 - 3)$

$x(x-\sqrt{3})(x+\sqrt{3})$



Domain $\rightarrow (-\infty, \infty)$

Range $\rightarrow (-\infty, \infty)$

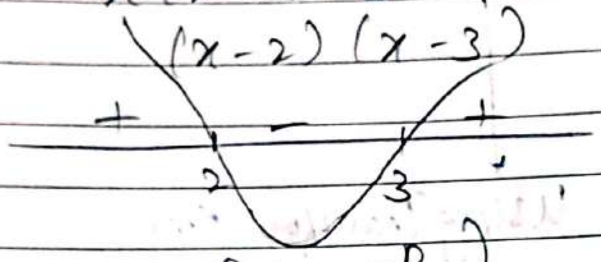
Q $y = x^2 - 5x + 6 \longrightarrow a = 1 > 0$

Parabola
2 distinct points

$x^2 - 2x - 3x + 6$

$x(x-2) - 3(x-2)$

$(x-2)(x-3)$



$\left(\frac{-b}{2a}, \frac{-D}{4a}\right)$

$a = 1,\ b = -5,\ c = 6$

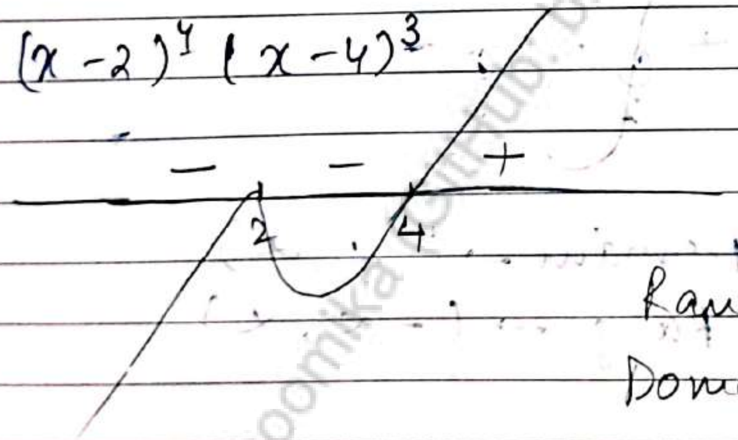
$y = \frac{-(b^2 - 4ac)}{4a}$

$= -1/4$

Range $[-1/4, \infty)$

Domain $(-\infty, \infty)$.

Q $(x-2)^4 (x-4)^3$



Range $(-\infty, \infty)$
Domain $(-\infty, \infty)$

Q $x^3 - 2x^2 + x = 0$

1	1	−2	1	0
	$\cancel{4}$	1	−1	0
	1	−1	0	0

Horner's Method

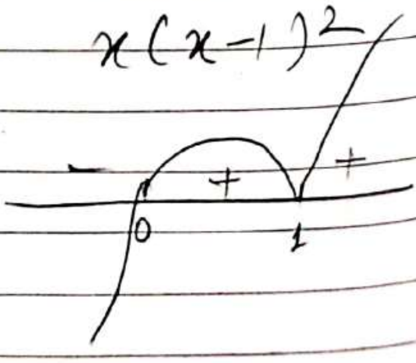
$(x-1)(x^2 - x) = 0$

$x(x-1)(x-1) = 0$

$x(x-1)^2 = 0.$

$x(x^2 - 2x + 1)$

$x(x-1)^2$



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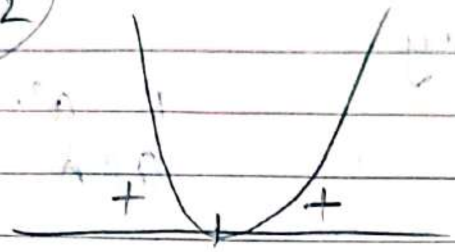
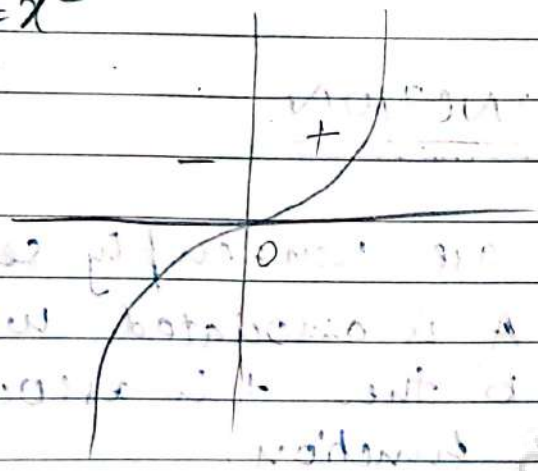
(1) $y = x^2$ even power



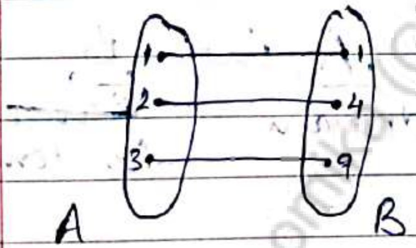
Domain $(-\infty, \infty)$
Range $\to [0, \infty)$

(1) $y = x^3$



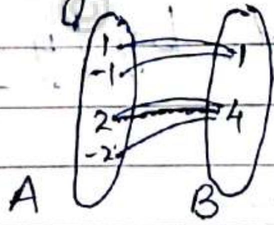
RELATIONS

(1) One – One



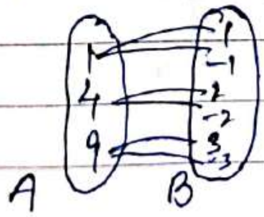
$R: a^2 = b$
$a \in A \quad b \in B$

(2) Many – One



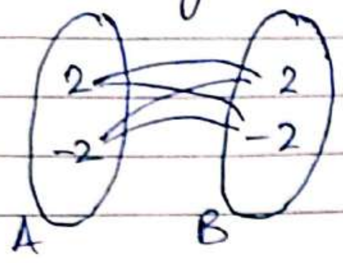
$R: a^2 = b$
$a \in A \quad b \in B$

(3) One – Many.



$R: a = b^2$
$a \in A, \ b \in B$

(4) Many - Many



R: $a^2 = b^2$
$a \in A \quad b \in B$

FUNCTION

Definition: If A & B are non-empty sets s.t each element of A is associated with unique element of B then this association is known as function.

(2) Function = Correspondence = Mapping

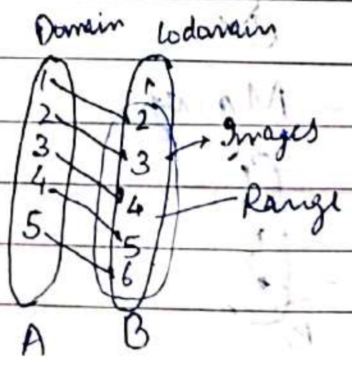
(3) Represented by :- $f : A \longrightarrow B$

$\underbrace{A}_{\text{domain}}$ into $\underbrace{B}_{\text{Co-domain}}$ ~~Range~~

(4) Domain = Input
Range = Output.

(5) Function is represented by $\boxed{y = f(x)}$

(I) $y = f(x) = x + 1$
$x \in A$
$y \in B$



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(1) Range $\subseteq$ Co-domain
(2) Range can never bee greater than co-domain
(3) Range may be equal to co-domain

$\Rightarrow$ Range = Set of all images
Domain = Set of all pre-image

y = Range $\qquad$ x = Domain

(II) Ordered Pair Method :-

$f = \{(1,2), (2,3), (3,4), (4,5), (5,6)\}$
$f^{-1} = \{(2,1), (3,2), (4,3), (5,6), (6,5)\}$

$f : A \rightarrow B$
$f^{-1} : B \rightarrow A$
But f^{-1} is not defined

(1) Image of 3 = 4
$f(3) = 4$
$f^{-1}(6) = 5$

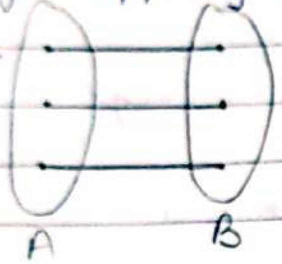
(1) Pre Image of 5 is 4

In Function :-
(I) There should be a 'y' for every 'x'
(II) There should be only one 'y' for every 'x'
(III) For every 'x' 'y' should be defined

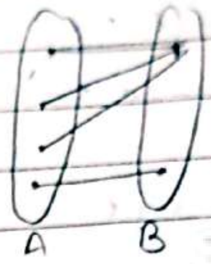
Q Find which are its function & which are 4 methods

(A) By mapping

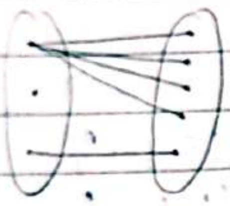
(1) Function



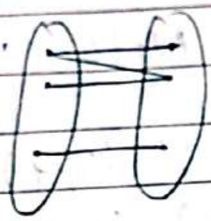
(2) ✓



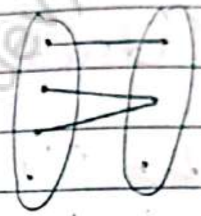
(3) ×



(4) ×



(5) ×



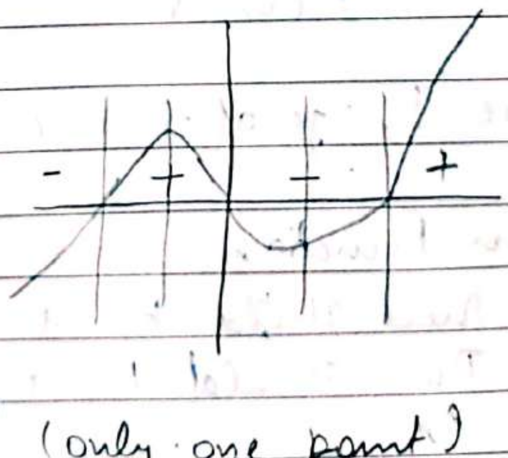
Vertical Line Test

If any vertical line cuts the graph of function at more than one point then its is not a function

(If line intersect at one point then its is function)

$y = x^3 - x$

Roots :- $x(x-1)(x+1)$



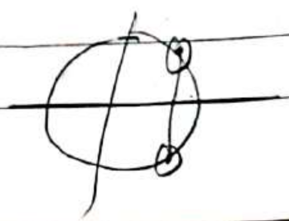
This function because vertically cut only one time or one point } (only one point)

$x^2 + y^2 = 4$

Not a function, & two points

$x^2 + y^2 = x^2$

$1 = 2$



Function

Algebraic
- Binomial fraction
- Rational fraction
- Irrational fraction
- Piecewise fraction
  - Modulus fraction
  - Sign fraction
  - GIF
  - fractional part fraction
  - defined fraction / Limit continuity

Transcedental
- Logarithmic
- Inverse
- Trigonometric
- Exponential

Ex. $f(x) = \begin{cases} x^3 & x < 0 \\ 3x-2 & 0 \le x \le 2 \\ x^2+1 & x > 2 \end{cases}$

Q. Find value of $f(-1) + f(1) + f(3)$?

(b) Find $f(x)$ if $x = 4$

As $f(-1) = x^3 \rightarrow (-1)^2$

$f(1) = 3(1) - 2$

$f(3) = 9 + 1 = 10.$

(b) $x^2 + 1$

Polynomial function :-

- $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x^1 + a_n$
- If $f(x) = a_{n-1} x + a_n \rightarrow$ Polynomial
- $f(x) = a_n \rightarrow$ Polynomial (constant polynomial)
- $f(x) = a_{n-2} x^2 + a_{n-1} x^1 + a_n \rightarrow$ Polynomial
- $f(x) = ax^2 + bx + c \rightarrow$ quadratic function

$f(x) = ax + b \rightarrow$ Linear function

$f(x) = a \rightarrow$ Constant function

Even polynomial → $ax^4+bx^3+cx^2+dx+e$
↳ degree → even
odd polynomial → odd

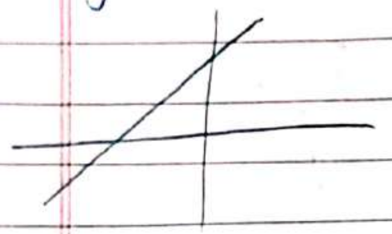
* Highest degree :- 'n' → in polynomial

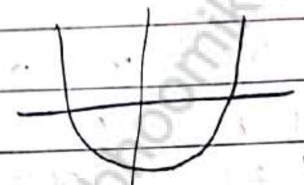
Range of odd degree polynomial is $\mathbb{R}$

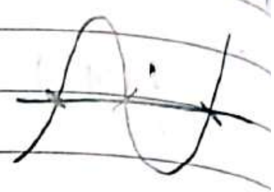
. $f(x)=x^5+x^3+7x^2+1$ → Range $=\mathbb{R}$

. $f(x)=x$ → Range $=\mathbb{R}$

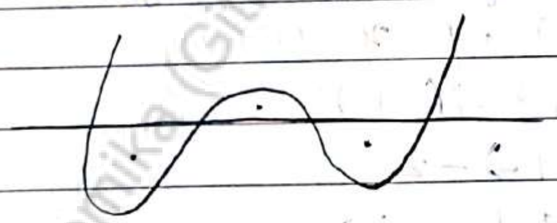
① $y=ax+b$ ② $y=ax^2+bx+c$ ③ $y=ax^3+bx^2+c$







④ $y=ax^4+bx^3+cx^2+dx+e$



$f(x)=ax^2+bx+c = a(x-\alpha)(x-\beta)$

α β → Roots

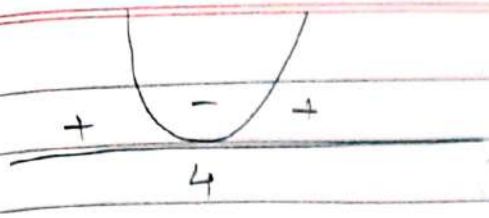
$f(x)=ax^3+bx^2+cx+d = a(x-\alpha)(x-\beta)(x-\gamma)$

α β γ

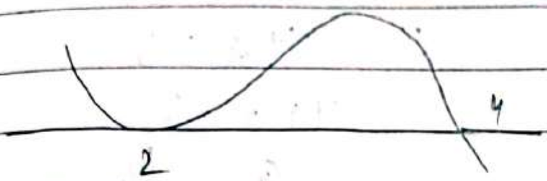
$f(x)=a_0x^n+a_1x^{n-1}+\ldots+a_{n-1}x+a_n$

α_1 α_2 α_3 .. α_n

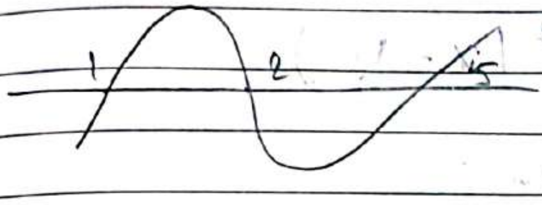
$a(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_n)$



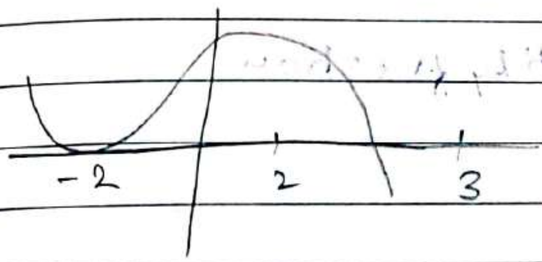
$f'(x) = a(x-4)^2$



$f(x) = a(x-2)^2(x-4)$



$f(x) = a(x-1)(x-2)$
$(x-5)$



Find $f(x)$ if $f'(x) = 3$
at B

$$\left.\frac{dy}{dx}\right|_B = 3$$

slope (tangent at Point $B = 3$)

$f(x) = a(x+2)^2(x-\alpha)$

$f(x) = a(x+2)^2(x-\alpha)$

$y + a(x+2)^2(x-\alpha)$

$5 = -4a\alpha$

$y = a(x+2)(x-\alpha)$

$5 = -4a\alpha$

$y = f(x)\,g(x)$

$\frac{dy}{dx} = f'(x)\,g(x) + f(x)\,g'(x)$

$f'(x) = a[(x+2)^2(1) + (x-\alpha)^2(x+2)]$

$= a[2^2 + (-2\alpha)(2)]$

$\Rightarrow 3 = a[1-4\alpha]$

$a[4 - 4(-5/4a)]$

$2(x-\alpha)$
$(x+\alpha)$
$2(-\alpha)(2)$
-4α

$$\alpha = -\frac{5}{4a}$$

$$= -\frac{5}{4} \cdot 2(-2)$$

$$\alpha = 5/2$$

$$3 = 4a + 5$$

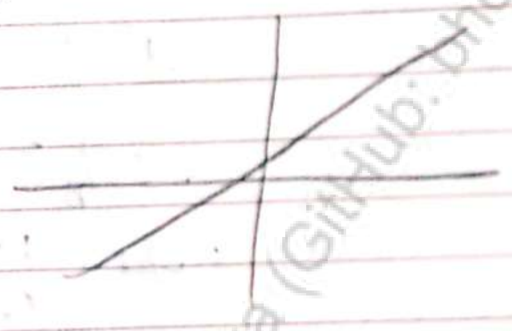
$$4a = -2$$

$$a = -1/2$$

$$f(x) = \frac{1}{2}(x+2)^2(x - 5/2)$$

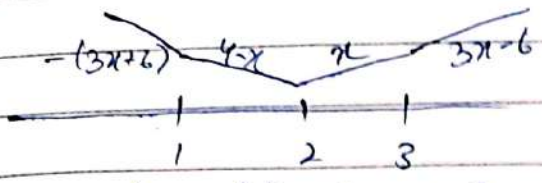
Identity fraction :-

(a) $y = x$ is identity fraction



Range $\mathbb{R}$
Domain $\mathbb{R}$

$|x-1|+|x-2|+|x-3) \rightarrow$ Identity function



$(x-1)-(x-2)-(x-3)$ $\qquad$ $x-1+x-2-(x-3)$

$-1+2-x+3$ $\qquad$ $2x-3-x+3$

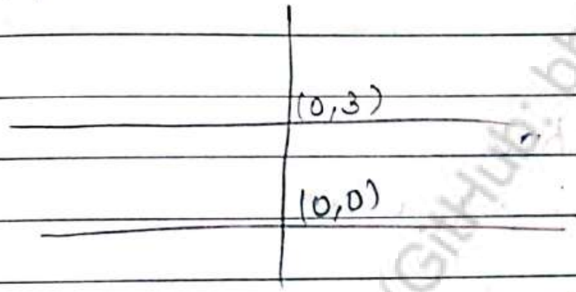
$4-x$ $\qquad$ x

$[2,3]$

$y=x$ satisfy

CONSTANT FUNCTION

ⓐ $f(x)=k$ is constant function



graph of $f(x)=3$ is line

Range $=\{3\}$

Domain $=(-\infty,\infty)$

$f(0)=3 \quad f(5)=3$

Q) $f(x) = \dfrac{\cos^2 x - \sin^4 x}{\cos^4 x + \sin^2 x}$ then $f(2026)=?$

Sol $\cos^2 x + \sin^4 x$] Numerator

$(\sin^2 x)^2$

$(1-\cos^2 x)^2$

$\cos^2 x + 1 + \cos^4 x - 2\cos^2 x$

$\boxed{\cos^4 x - \cos^2 x + 1}$

$\cos^4 x + \sin^4 x$

$f(x)=1$

$f(2026)=1$

① $f(x) = |x-1| + |x-5|$

$(2x-6)$ —— 4 —— $2x-6$

1 —— 5 (minus ke do)

$1-5 \rightarrow$ 4 ayega. constant ayega.

$|5-1| = 4$

SPECIAL CONSTANT FUNCTION

Result 1 $f(x) = \sin^2 x + \sin^2(x + \pi/3) + \cos x(\cos(x + \pi/3)) = 5/4$

$g(5/4) = 1$, then $gof(x) = 1$

Result 2 $\cos^2 x + \cos^2(x + \pi/3) - \cos x(\cos(x+\pi/3)) = 3/4$

Exponential function.

$f(x) = a^x$ ($x \rightarrow$ variable, $a \rightarrow$ constant)

$a > 0$, $a \neq 1$

$a < 0 \rightarrow$ This is not a function

2^x, $\left(\frac{1}{2}\right)^{2x}$ $(\sin x)^x$ i.e $(\text{variable})^{\text{variable}}$

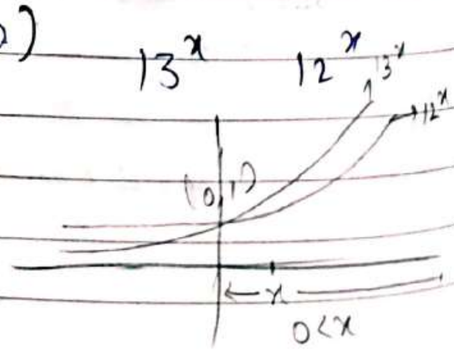
$\therefore \boxed{\sin x > 0}$

$x^x \rightarrow x > 0 \rightarrow (0, \infty)$

$13^x > 12^x \rightarrow x > 0$

Power pe depend karta hai

$\boxed{12^x > 13^x} \rightarrow x < 0$



$a^{-1} = \frac{1}{a}$, $a^2 = a \cdot a$

*
$13^x > 12^x$ Domain $\in (0, \infty)$
$12^x > 13^x$ Domain $(-\infty, 0)$

Q. Domain of $\sqrt{2^x - 5^x}$

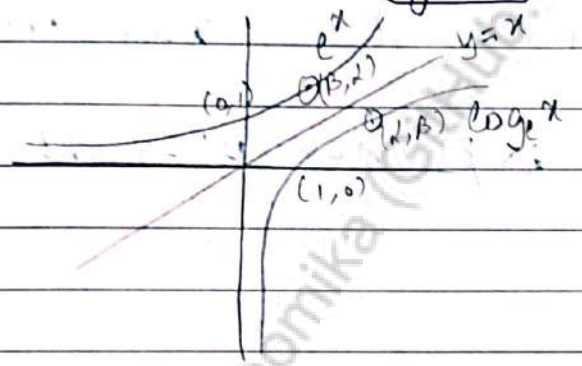
Sol $f(x) \geq 0$

$2^x - 5^x \geq 0$

$2^x \geq 5^x$

$x \in (-\infty, 0]$

e^x & $\log_e x$ are inverse fxn to each other
Inverse function are mirror image of each other in line $\boxed{y = x}$

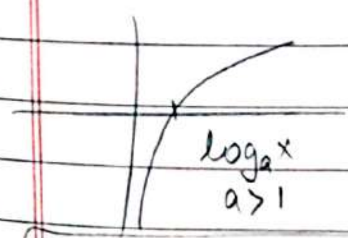


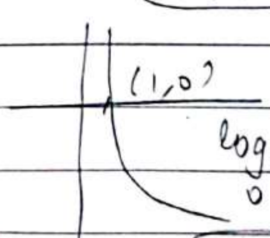
$y = \log_e x$
$x = e^y$

$y = \log_a x$
* = +ve

$\boxed{\log_a 1 = 0}$

$a > 1$ & $x > 1$ $0 < a < 1$ & $x < 1$





* -ve $\boxed{a > 1 \text{ \& } 0 < x < 1}$ $\boxed{0 < a < 1 \text{ \& } x > 1}$

$\log_{.2} .4 \rightarrow +ve$	Dono negtive → positive
$\log_{.2} 4 \rightarrow -ve$	argument +ve dusra negative → -ve
$\log_{2} .4 \rightarrow -ve$	base +ve a -ve → -ve
$\log_{2} 4 \rightarrow +ve$	dono positive → positive

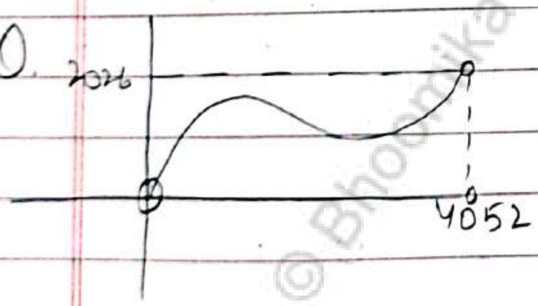
DOMAIN AND RANGE

Recap

Domain :- ① values of x for fxn is defined
② Domain is expression of graph on x-axis
③ Set of pre-images

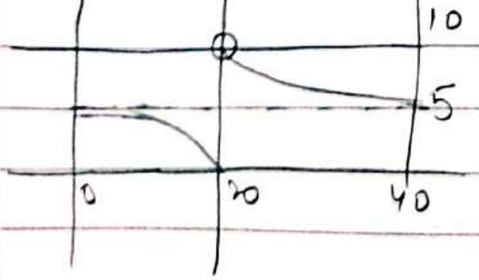
Range :- ① For valid 'x' value of y are range
② Range = y = height = answer = Image
③ Set of Images
④ Range as expression of graph on y-axis

Q.



Sol
Domain = $(0, 452)$
Range = $(0, 2026)$

Q.



Sol.
Domain = $[0, 40]$
Range = $[0, 10) - \{5\}$

③ Rules of Domain

(i) $\sqrt{f(x)} \longrightarrow f(x) \geq 0$

(ii) $\dfrac{1}{f(x)} \longrightarrow f(x) \neq 0$

(iii) $\dfrac{1}{\sqrt{f(x)}} \longrightarrow f(x) > 0$

$\sqrt{5x-10}$,	$\dfrac{1}{5x-10}$
Domain	Domain
$5x-10 \geq 0$	$\mathbb{R}-\{2\}$
$5x \geq 10$	$(-\infty,2)$
$x \geq 2$	
$[2,\infty$	

④ $f+g$
$f-g$
$f \times g$
$D_f \cap D_g$.

Q Domain of $\sqrt{x-6}+\sqrt{4-x}$

$x-6 \geq 0 \quad 4-x \geq 0$

$x \geq 6 \quad x \leq 4$

$D_f\ [6,\infty) \cap (-\infty,4]$

ϕ

Q $\sqrt{x-4}+\sqrt{6-x}$

$x-4 \geq 0 \quad 6-x \geq 0$

$x \geq 4 \quad x \geq 6$

$D_f\ [4,\infty) \cap [-6,\infty)$.

$x \in [4,6]$

Q $y=(x^3-x^2+x+1)\sqrt{x^2-4}$

$(-\infty,2] \cup [2,\infty)$

$x^2-4 \geq 0$

$(x+2)(x-2) \geq 0$

$+ \quad - \quad +$

$-2 \quad 2$

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Recap

Domain # $\sin^{-1}(f(x))$

$$-1 \le f(x) \le 1$$

$\log_{g(x)} f(x)$

$f(x) > 0$ & $g(x) > 0$ & $g(x) \neq 1$

$$\underbrace{D_f \cap D_g}$$

Signum function

☆ 'R club' → 2^x 3^x $\left(\frac{1}{2}\right)^x$, constant, fractional part, polynomial, $|x|$

$\sin x$, $\cos x$, $\tan^{-1}x$, $\cot^{-1}x$, greatest Integer $[x]$.

2^{x^2} → Domain of x^2 $\qquad 2^{\sin^{-1}x}$ → Domain of $\sin^{-1}x$

Concept of 'R-club'

In Ques. of member of R-club, we always say the domain of function, which is sitting at ~~to~~ the place of x

Q $y = (\sin^{-1}x)^3 - 3(\sin^{-1}x) + 2$

$\underbrace{x^3 - 3x + 2}_{\text{polynomial}}$ [R club ka member]

Domain of $\sin^{-1}x$ //

$[-1, 1]$

Q. $y = 2^{\sin^{-1}\left(\frac{2-|x|}{4}\right) + \cos^{-1}\left(\frac{2-|x|}{4}\right)}$

Domain = $\sin^{-1}\left(\frac{2-|x|}{4}\right) + \cos^{-1}\left(\frac{2-|x|}{4}\right)$

$$-1 \le \frac{2-|x|}{4} \le 1$$

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$-4 \leq 2-|x| \leq 4$

$-6 \leq -|x| \leq 2$

$6 \geq \underline{|x| \geq -2}$

Ignore.

$|x| \leq 6$

$[-6, 6]$

① $y = 2^{\sin^{-1}(\log_2 x)}$

$-1 \leq (\log_2 x) \leq 1$

$2^{-1} \leq x \leq 2^{1}$

$\frac{1}{2} \leq x \leq 2 \qquad \left[\frac{1}{2}, 2\right]$

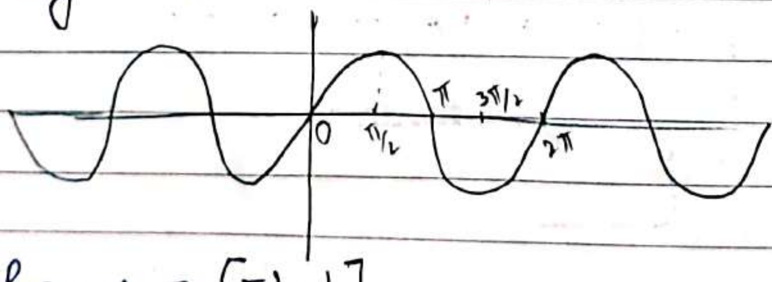
① $\sqrt{e^{\sin^{-1}(\log_2 x)}}$

$e^{\sin^{-1}(\log_2 x)} \geq 0$

$\left[\frac{1}{2}, 2\right]$

TRIGNOMETRIC FUNCTION

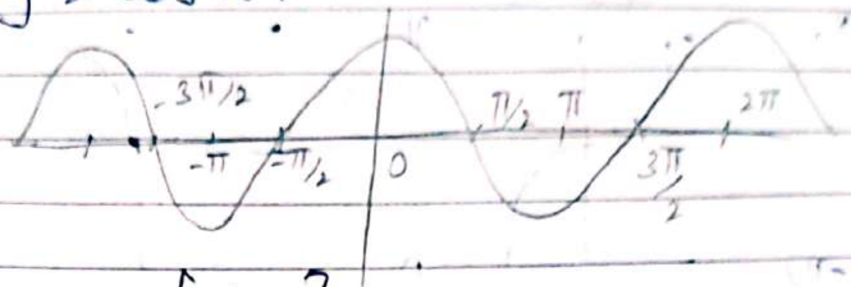
* $y = \sin x$



① Range = $[-1, 1]$
② Domain $\mathbb{R}$
③ $\sin x = 0 \Rightarrow \boxed{x = n\pi}$
④ $\sin x > 0 \Rightarrow \boxed{2n\pi + 0 < x < 2n\pi + \pi}$

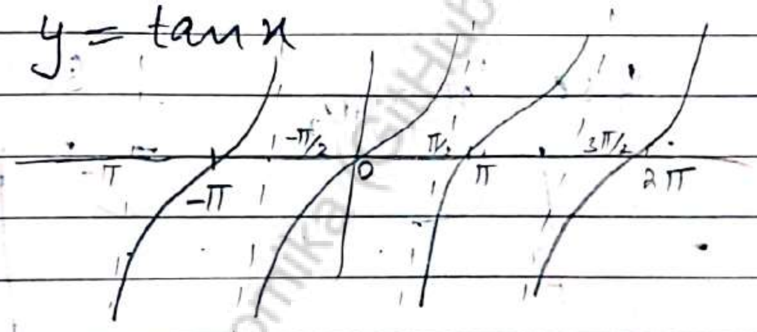
5) $\sin x < 0 \Rightarrow \boxed{2n\pi + \pi < x < 2n\pi + 2\pi}$

☆ $y = \cos x$.



1) Range $= [-1, 1]$
2) Domain $= \mathbb{R}$
3) $\cos x = 0 \Rightarrow \boxed{x = (2n+1)\frac{\pi}{2}}$ odd multipl
4) $\cos x > 0 \Rightarrow \boxed{2n\pi - \frac{\pi}{2} < x < 2n\pi + \frac{\pi}{2}}$
5) $\cos x < 0 \Rightarrow \boxed{2n\pi + \frac{\pi}{2} < x < 2n\pi + \frac{3\pi}{2}}$

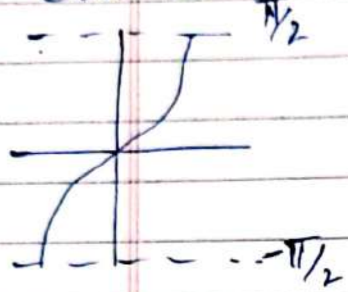
☆ $y = \tan x$



1) Range $= \mathbb{R}$
2) Domain $=$ ~~(2n+1)π/2~~ $R - (2n+1)\pi/2$
3) $\tan x = 0$ $\boxed{x = n\pi}$
4) $\tan x > 0$ $\boxed{n\pi + 0 < x < \pi/2 + n\pi}$
5) $\tan x < 0$ $\boxed{n\pi + \pi/2 < x < n\pi + \pi}$

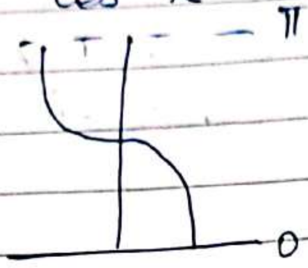
INVERSE TRIGNOMETRIC FUNCTION.

$\sin^{-1}x$



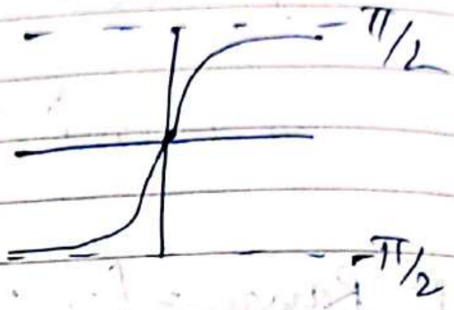
Domain
$[-1, 1]$
Range
$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\cos^{-1}x$



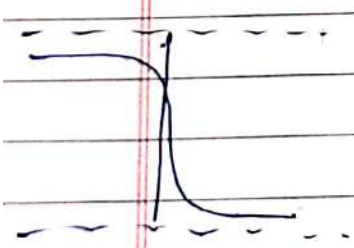
Domain
$[-1, 1]$
Range
$[0, \pi]$

$\tan^{-1}x$.



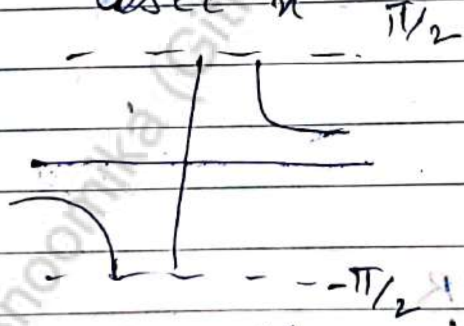
Domain
$\mathbb{R}$
Range
$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$\cot^{-1}x$.



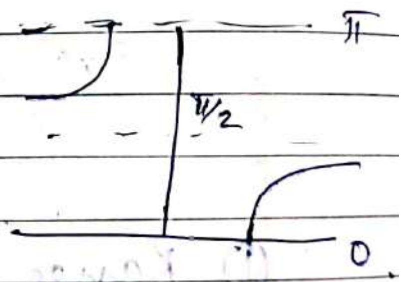
Domain
$\mathbb{R}$
Range
$(0, \pi)$

$\operatorname{cosec}^{-1}x$



Domain
$(-\infty, -1] \cup [1, \infty)$
Range
$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

$\sec^{-1}x$



Domain
$(-\infty, -1] \cup [1, \infty)$
Range
$[0, \pi] - \{\frac{\pi}{2}\}$

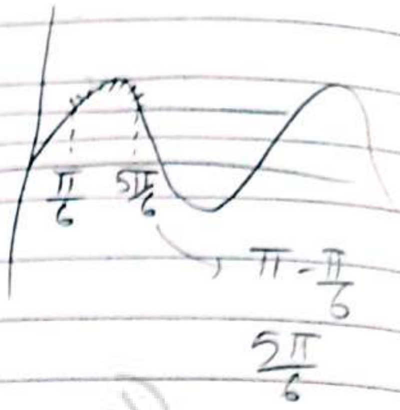
⑤ $\dfrac{1}{\sqrt{2\sin x - 1}}$

Sol

$2\sin x - 1 > 0$

$\sin x > 1/2$

$\dfrac{\pi}{6} < x < \dfrac{5\pi}{6}.$



GREATEST INTEGER FUNCTION

① Represented by $f(x) = [x]$

② $[x] \rightarrow$ takes its left side integer value

$[3.89] = 3$

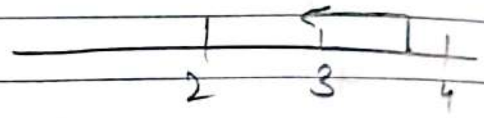
$[4.12] = 4$

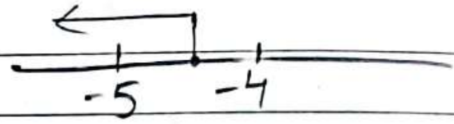
$[9] = 9.$

$[-4.12] = [-5]$

$[-0.12] = [-1]$

$\left[-\dfrac{1}{3}\right] = -1.$





Ques. $f(x) = \sin[\pi^2]x + \sin[-\pi^2]x$

then $f(\pi/2) = ?$

Sol $f(x) = \sin[9.86]x + \sin[-9.86]x$ $\quad \pi \rightarrow 3.14$, $\pi^2 \rightarrow 9.86$

$\sin[9]x + \sin[-10]x$

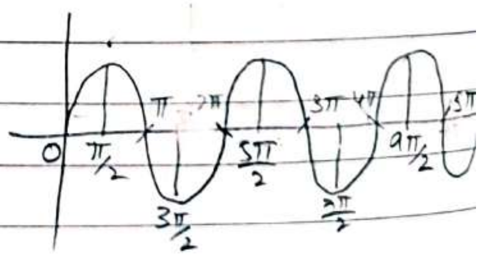
$f(x) = \sin 9x - \sin 10x$

$f(\pi/2) = \sin 9\dfrac{\pi}{2} - \sin \cancel{10}^{5} \dfrac{\pi}{\cancel{2}}$

$= \sin 9\dfrac{\pi}{2} - \sin 5\pi$

$= 1 - 0$

$= 1$ Ans



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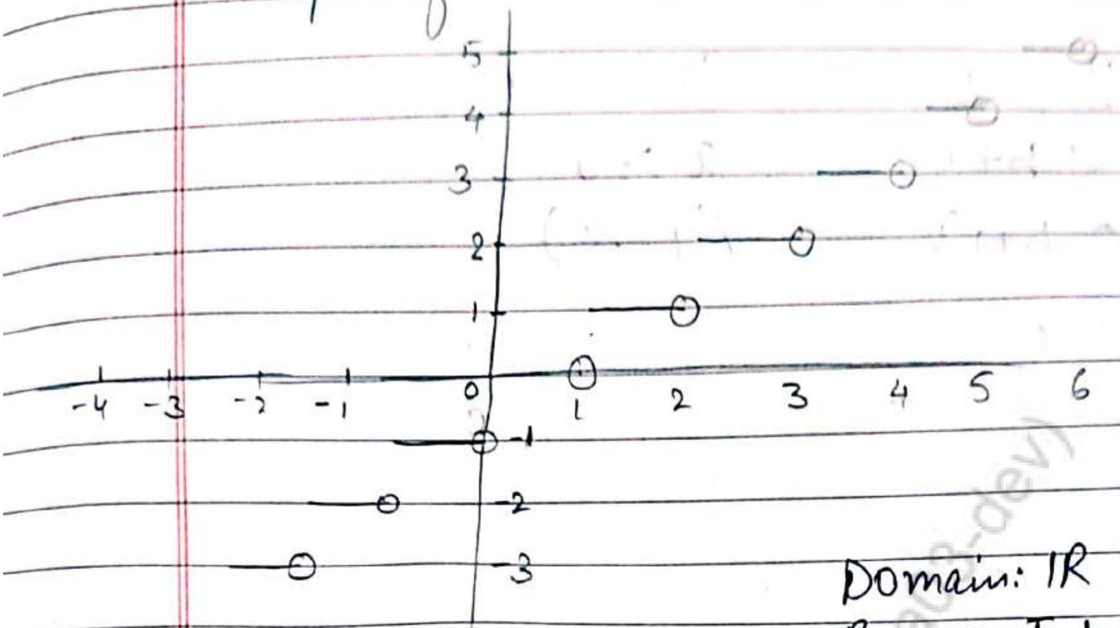
Graph of $[x]$

{ 0 se shuru karna hai }



Domain: $\mathbb{R}$
Range: Integer (I)

PROPERTIES

★ $[I] = I$

★ $[-I] = -[I] = -I$ eg $[-9] \neq -10$

★ $[x+I] = [x] + I$

★ $[x] = m \Rightarrow m \le x < m+1$

eg.

$[x] = 4$ $[x] = -9$

$4 \le x < 5$ $-9 \le x < -8$

Ques $[x + [x + [x + [x]]]] = 4$

$x = ?$

Sol $[1, 2)$

$[x]+[x]+[x]+[x] = 4$

$4[x] = 4$

$[x] = 1$

$1 \le x < 2$

★ $[x] + [-x] = \begin{cases} 0 & x \in I \\ -1 & x \notin I \end{cases}$

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★ $x \le [x] < x+1$ $\quad 3 \le [3.4] < 4$

★ $[x] \le b$ $\qquad [x] \ge a$

$x < b+1$ $\qquad x \ge a$

$(-\infty, b+1)$ $\qquad [a, \infty)$

Ques $3 \le [x] \le 5$

$x \le 5+1$

$x < 6$.

$[3, 6)$

Ques $-1 \le [x] < 2$

$[-1, 2)$

Ques. $-1 < [x] < 2$ $\rightarrow -1$

0

1

$[-1, 0) \cup [0, 1) \cup [1, 2)$

Ques $-2 \le [x] < 1$

$[x] = -2 \rightarrow -2 \le x < 1$

$[x] = -1 \rightarrow -1 \le x < 0$

$[x] = 0 \rightarrow 0 \le x < 1$

$[-2, 1)$

Ques. $-3 < [x] < -1$

$[x] = -2 \longrightarrow -2 \le x < 1$

Ques $-1 \le [x] \le 3$

$[x] = -1$ $\quad -1 \le x < 0$ $\qquad [x] = 2$ $\quad 2 \le x < 3$

$[x] = 0$ $\quad 0 \le x < 1$ $\qquad [x] = 3$ $\quad 3 \le x < 4$

$[x] = 1$ $\quad 1 \le x < 2$

$[-1, 4)$

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Q. $y = 2[x] + 3$, $y = 3[x-2] + 5$ then $[x+y] =$

sol $y = 2[x] + 3$ $\quad$ $y = 3[x] - 6 + 5$

$y = 3[x] - 1$

$$y = 2x + 3$$
$$y = 3x - 1$$
$$-x + 4$$
$$x = 4$$

$y = 8 + 3$
$= 11$

$$\underbrace{[x+y]}_{[x]+11} = 15$$

Q. $y = \sqrt{[x]^2 - 5[x] + 6}$

$$[x]^2 - 5[x] + 6 \geq 0$$
$$[x]^2 - 3[x] - 2[x] + 6 \geq 0$$
$$[x]([x]-3) - 2([x]-3) \geq 0$$
$$([x]-2)([x]-3) \geq 0$$

+ $\quad$ − $\quad$ +
2 $\quad$ 3

PROPERTY (I) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \dots \left[x + \frac{n-1}{n}\right] = [nx]$

Ques $\left[\frac{1}{4}\right] + \left[\frac{1}{4} + \frac{1}{100}\right] + \left[\frac{1}{4} + \frac{2}{100}\right] + \dots + \left[\frac{1}{4} + \frac{99}{100}\right]$

Ans $\left[\frac{1}{4} \times 100\right] = 25$

Ques. $[x] + \left[x + \frac{1}{100}\right] + \cdots \left[x + \frac{99}{100}\right] = 7$

find $x = ?$

Ans. $[100x] = 7$

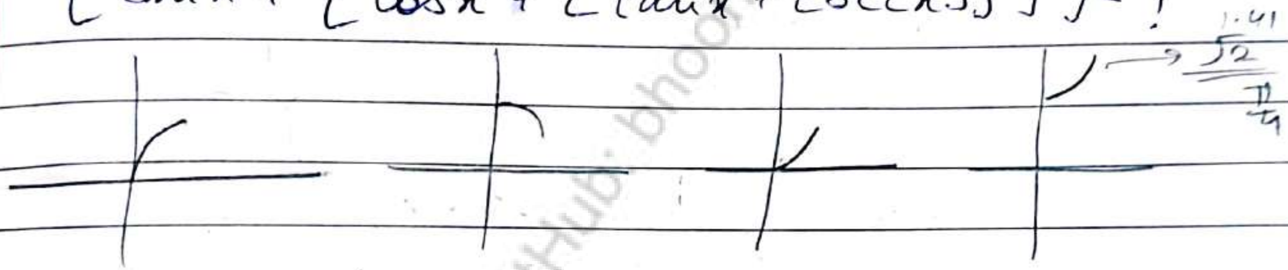
$7 \le 100x < 8$

$\frac{7}{100} \le x < \frac{8}{100}$

$x \in \left[\frac{7}{100}, \frac{8}{100}\right)$

Ques If $x \in (0, \pi/4)$

$[\sin x + [\cos x + [\tan x + [\sec x]]]] = ?$



$[\sin x] + [\cos x] + [\tan x] + [\sec x]$

$0 + 0 + 0 + 1$

1 Ans.

(II) $[x] + \left[\frac{1}{x}\right] = 2$

$[x]$ → Integer, $\left[\frac{1}{x}\right]$ → Integer

1	1
0	2
2	0

Case (I) $[x] = 0$ $\quad \left[\frac{1}{x}\right] = 2$

$[0, 1)$ $\quad 2 \le \frac{1}{x} < 3$

$\frac{1}{2} \ge x > \frac{1}{3}$

$\boxed{\frac{1}{3} < x \le \frac{1}{2}}$

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Case (II) $[x] = 1$ $\quad [1/x] = 1$

$1 \leq x < 2$ $\quad 1 \leq \frac{1}{x} < 2$

$1 \geq x > \frac{1}{2}$

$[1, 2) \cap [1/2, 1]$

$\boxed{x = 1}$

Case (III) $[x] = 2$ & $[1/x] = 0$

$2 \leq x < 3$ $\quad 0 < \frac{1}{x} < 1$

$\infty > x > 1$

$[2, 3) \cap (1, \infty)$

$\boxed{[2, 3)}$

// $x \in [\frac{1}{3}, \frac{1}{2}] \cup \{1\} \cup [2, 3)$

FRACTIONAL PART

1. Represented by $f(x) = \{x\}$
2. It show after decimal value {fractional value}
3. It is always ~~positive~~ [Non-negative]

$\{3.14\} = 0.14$

$\{3.23\} = 0.23$

$\{0.001\} = .001$

$\{-1.0\} = .0$

$\{0\} = .0$

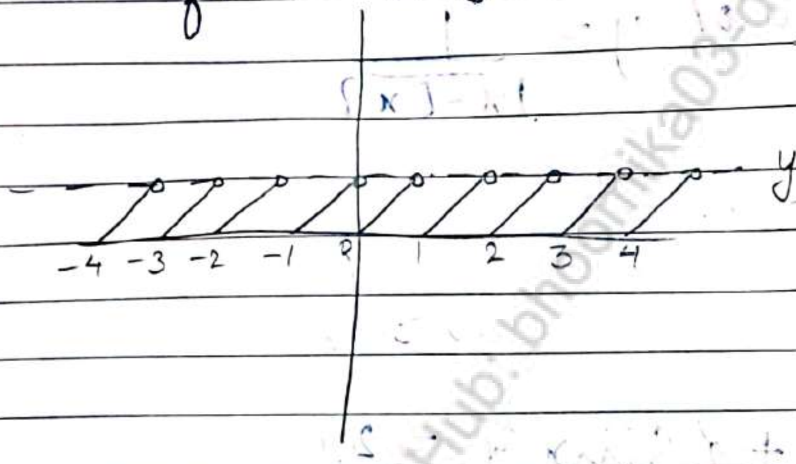
$\{-1.24\} = 1 - 0.24$

$\{-9.86\} = 1 - 0.86$

$\{\pi\} =$ Not defined, $\{e\} =$ Not define

④ Graph of fractional part

$f(x) = \{x\}$



Domain $\mathbb{R}$.

Range $[0, 1)$

$0 \le y < 1$

Ques $y = \{\sin^{-1}x\}$

Domain of $\sin^{-1}x$ $[-1, 1]$

① PROPERTY

(i) $0 \le \{x\} < 1$

(ii) $\{x\} = 0 \Rightarrow x =$ Integer

$\{x\} \ge 0$ $\{x\} > 0 \Rightarrow \mathbb{R} -$ Integer

eg $\{1\} = .01$

$\{1.1\} = 0.1$

(iii) $\{x + n\} = \{x\}$

(iv) $\{x\} + \{-x\} = \begin{cases} 0 & x \in I \\ 1 & x \notin I \end{cases}$

Range of $y = x - [x-2]$

$y = x - [x] + 2$

$\{x\} + 2$

$\boxed{0 \le \{x\} < 1}$ $\quad y \in [2,3)$

$2 \le \{x\} + 2 < 3$

$2 \le y < 3$

① Domain of $y = \sqrt{\dfrac{\sqrt{4-x^2}}{2\{x\}-1}}$

Sol

$$\frac{\sqrt{4-x^2}}{2\{x\}-1} \ge 0$$

$2\{x\} - 1 > 0$

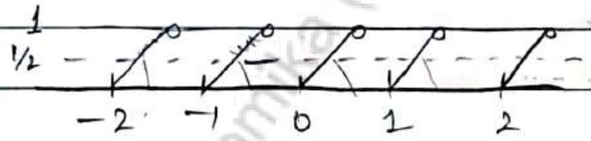
$\{x\} > 1/2$

$4 - x^2 \ge 0$

$(2-x)(x+2) \ge 0$

$(x-2)(x+2) \le 0$

$-2 \le x \le 2$



$\left(-\frac{3}{2}, -1\right) \cup \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, 1\right) \cup \left(\frac{3}{2}, 2\right)$

SIGNUM FUNCTION

① Represented by $y = \mathrm{Sgn}(x)$

② $\mathrm{Sgn}(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \\ 0 & x = 0. \end{cases}$

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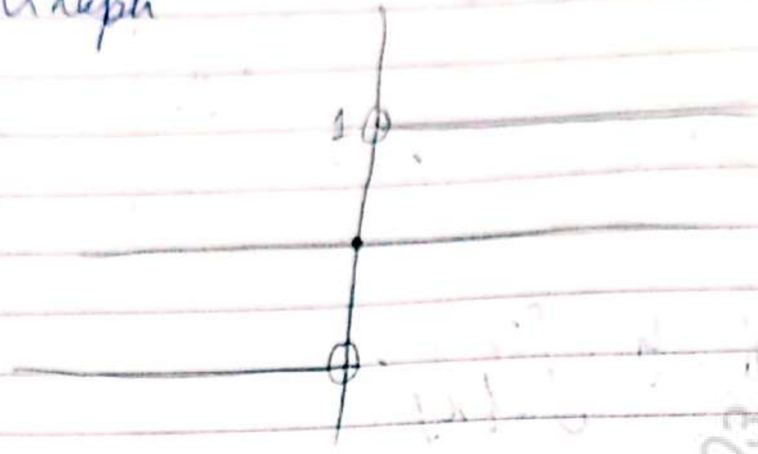
Range $= \{-1, 0, 1\}$ Domain $= \mathbb{R}$

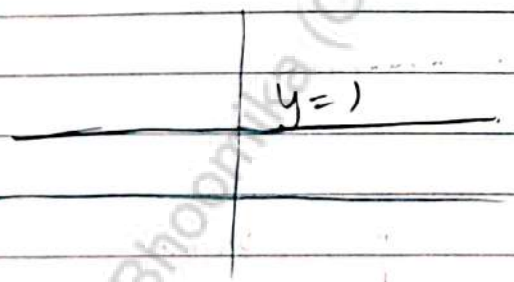
(4) Graph



$\operatorname{Sgn} f(x) = \begin{cases} 1 & f(x) > 0 \\ -1 & f(x) < 0 \\ 0 & f(x) = 0 \end{cases}$

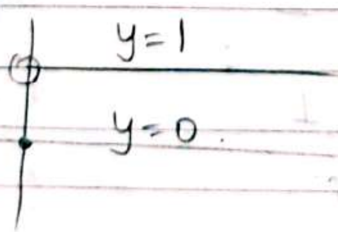
Q. $\operatorname{Sgn}(e^x) = \{ 1 \quad e^x > 0$.

$\forall x \in \mathbb{R}$



Q $\operatorname{Sgn}(x^2) = \begin{cases} 1 & x^2 > 0 \\ -1 & x^2 < 0 \\ 0 & x^2 = 0 \end{cases}$

$\{0, 1\}$.



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COMPOSITION OF FUNCTION

Def Let $f: X \to Y$ & $g: Y \to Z$ be two function with D_f & D_g respectively then composite function of f & g is written as $gof: X \to Z$ & defined as $gof(x) = g(f(x))$

Similarly
$fog: Z \to X$ & defined as $f(g(x))$

Domain of $(fog) = \{x : x \in D_f \;\&\; x \in D_g\}$
&.
Domain of $gof = \{x : x \in D_g \;\&\; x \in D_f\}$

Example $f(x) = \sin x$ $\quad g(x) = \cos x$

$fog(x) = \sin(\cos x)$ $\qquad gof(x) = \cos(\sin x)$

REMARK
gof exists iff $R_f \subset D_g$.
fog exists iff $R_g \subseteq D_f$

Ques
Let $f(x) = [x]$ $\qquad g(x) = |x|$
then the value of $(gof)(-5/3) - fog(-5/3) =$

Sol $gof = g(f(x)) = g(f(-5/3)) = g(-2)$.

$f(-5/3) = \left[\frac{-5}{3}\right] = [-1.\text{som}] = -2$

$g(-2) = |-2| = 2$.

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$fog(x) = \sin|x|$ ✗

$gof(x) = (\sin\sqrt{x})^2$ ✗

$\sqrt{x^2} \longleftrightarrow |x|$

Not equal to given condition

(i) $g(f(x)) = \sqrt{\sin^2 x} = |\sin x|$

$f(g(x)) = \sin^2\sqrt{x} = (\sin\sqrt{x})^2$

(ii) $gof = |\sin x|$

$fog = \sin|x| \neq |\sin x|$

(iii) $gof = \sin\sqrt{x^2} = \sin|x| \neq \sin x$

RANGE

Range = y = height = Answer = Image

1. $[x]$ g.I.f $\rightarrow$ Range = I
2. $\{x\}$ $\xrightarrow{\text{fraction part}}$ Range = $[0, 1)$
3. $|x| \longrightarrow$ Range = $[0, \infty)$
4. $\log_e x \longrightarrow$ Range = $(-\infty, \infty)$ $\mathbb{R}$
5. $e^x, 2^x, 3^{-x} \longrightarrow$ Range = $(0, \infty)$
6. $y = \text{Sgn}\,x \longrightarrow$ Range = $\{-1, 0, 1\}$
7. $y = \text{Sgn}(+ve\ f^n) \longrightarrow$ Range $\{1\}$
8. $y = \text{Sgn}(e^{2x} + e^x + 1) \longrightarrow$ Range $\{1\}$
9. $y = \text{Sgn}(e^{2x} + e^x - 1) \rightarrow$ Range $\{-1, 0, 1\}$
10. $y = 4 \rightarrow$ Range = $\{4\}$
11. $y = x! \rightarrow$ Range = $\{1, 2, 6, 24, 120, 720, 5040 \ldots\}$

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Trigonometric function.

12. $-1 \le \sin x \le 1$

13. $-1 \le \cos x \le 1$

14. $-\infty < \tan x < \infty$

15. $-\infty < \cot x < \infty$

16. $\sec x \in \mathbb{R} - (-1, 1)$
$\operatorname{cosec} x \in \mathbb{R} - (-1, 1)$

17. $-1 \le \sin(ax \pm b) \le 1 \rightarrow$ Range yehi Rahegi andr effect nahi karega.

18. $-1 \le \cos(ax \pm b) \le 1$

19. $-a \le a\sin(x) \le a$

H.W # $y = \sqrt{(1-\sin x)\sqrt{(1-\sin x)\sqrt{(1-\sin x)} \cdots \infty}}$

Range of $y = \dfrac{1}{9\sin x - 7}$

$-9 \le 9\sin x \le 9$

$-16 \le 9\sin x - 7 \le 2$

$-\dfrac{1}{16} \ge \dfrac{1}{9\sin x - 7} \ge \dfrac{1}{2}$

$(-\infty, -1/16] \cup [1/2, \infty)$

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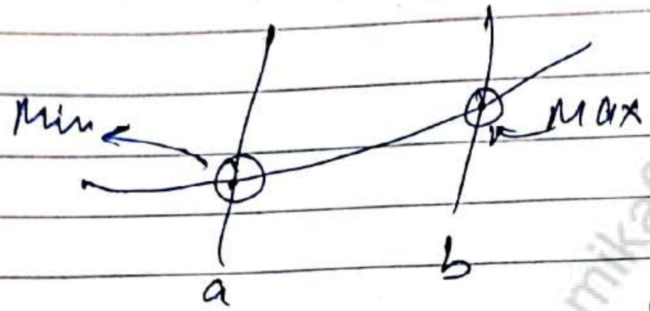
RANGE OF FINITE SET

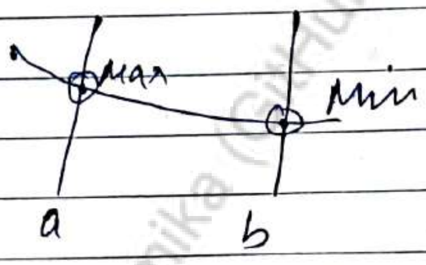
① If domain of function is a finite set, then check whether behaviour of function is increasing or decreasing

② If domain of $f(x)$ is $[a, b]$ & function is increasing



$$\text{Range} = [f(a), f(b)].$$

If function is decreasing



$$\text{Range} = [f(b), f(a)].$$

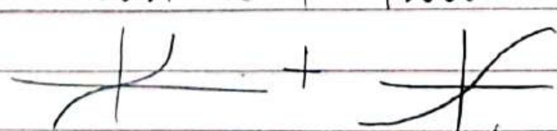
Q. $y = \frac{1}{\pi}[\sin^{-1} x + \tan^{-1} x]$ Range

① Find Domain

$$[-1, 1] \cap \mathbb{R} = [-1, 1]$$

② Behaviour

$\sin^{-1} x + \tan^{-1} x$



Increasing (overall)

$$\text{Range} = \left[\frac{1}{\pi}\left[\sin^{-1}(-1) + \tan^{-1}(-1)\right], \frac{1}{\pi}\left(\sin^{-1}(1) + \tan^{-1}(1)\right) \right]$$

$$\left[-\frac{\pi}{2} - \frac{\pi}{4}, \frac{1}{\pi}\left(\frac{\pi}{2} + \frac{\pi}{4}\right) \right]$$

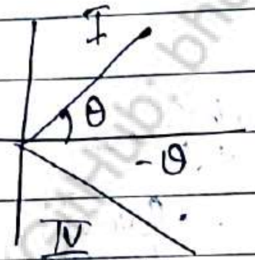
$$\left[-3/4, 3/4 \right]$$

EVEN FUNCTION AND ODD FUNCTION

Even function $\rightarrow$ $f(-x) = f(x)$ $\longrightarrow$ $y = x^2$ $\quad (-x)^2 = x^2$

$f(x) - f(-x) = 0$ $\quad y = |x|$ $\quad |-x| = |x|$

$y = \cos x$

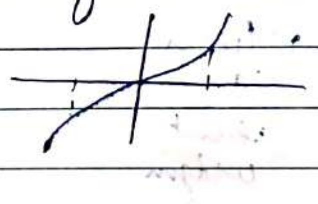


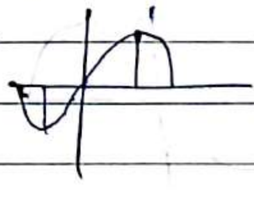
$\cos(-\theta) = \cos\theta$.

$\rightarrow$ ODD FUNCTION

$f(-x) = -f(x)$ $\rightarrow$ $y = x^3$, $y = \sin x$, $y = \tan x$ etc

$f(x) + f(-x) = 0$





Neither even nor odd function (NENO) :

$f(-x) \neq f(x)$
$\neq -f(x)$

$f(x) = 1 + \sin x$
$f(-x) = 1 - \sin x$
$\neq -f(x)$
$\neq f(x)$

$$f(x) = 1 + a^x$$
$$f(-x) = 1 + a^{-x}$$
$$\neq f(x)$$
$$\neq -f(x)$$

$$f(x) = a^x$$
$$f(-x) = a^{-x}$$
$$\neq -a^x$$
$$\neq a^x$$

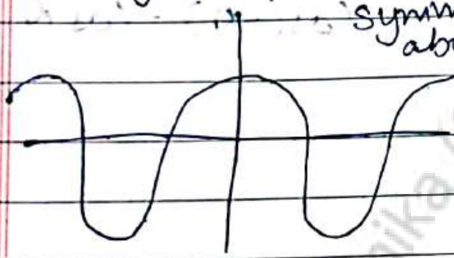
ON THE BASIS OF GRAPH

① Graph of even function is symmetric to y-axis

② Graph of odd function is symmetric to origin

$\boxed{\text{EVEN}}$

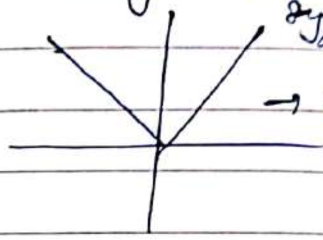
$y = \cos x$



symmetric about y-axis

→ even function

$y = |x|$

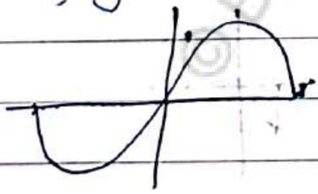


symmetric about y-axis

→ even function

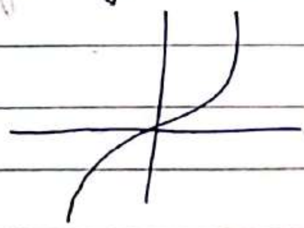
$\boxed{\text{ODD}}$

$y = \sin x$



→ turn 180° about origin

$y = x^3$



Note: Odd function is not necessary to touch origin

$$y = \mathrm{sgn}(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \\ 0 & x = 0 \end{cases}$$

$$\mathrm{sgn}(-x) = \begin{cases} -1 & x > 0 \\ 1 & x < 0 \\ 0 & x = 0 \end{cases}$$

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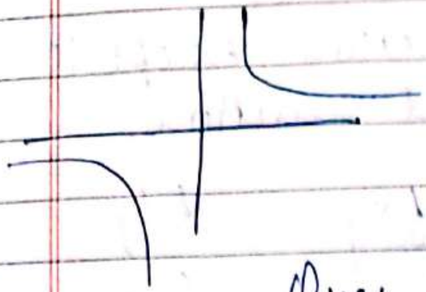
$y = \frac{1}{x}$ $\qquad y = \frac{1}{-x} = \frac{-1}{x}$



Symmetric about origin → odd function

Ques $h(x) = f(x) + f(-x)$
$h(x) = f(x) - f(-x)$
$h(x) = \frac{a^x - 1}{a^x + 1}$ check even/odd.

Sol ① Even
② Odd.
③ $h(-x) = \frac{a^{-x} - 1}{a^{-x} + 1} = \frac{\frac{1}{a^x} - 1}{\frac{1}{a^x} + 1} = \frac{1 - a^x}{1 + a^x} = -h(x)$

∴ odd function

Ques. $\left(\frac{a^x - 1}{a^x + 1}\right) \cdot x \rightarrow$ odd.
odd

$f(-x) = -\left(\frac{a^x - 1}{a^x + 1}\right)(-x)$

even function

Ques. $\left(\frac{a^x - 1}{a^x + 1}\right) \cdot x^2 \rightarrow$ even
odd
= odd.

Ques. $f(x) = \left(\frac{a^x - 1}{a^x + 1}\right) \cdot x^n$

For this n needs to be even

$n = \frac{2}{3}$ $\qquad x^n = x^{2/3} = (x^2)^{1/3}$
$f(-x) = ((-x)^2)^{1/3} = (x)^{1/3} \rightarrow$ even
or
$(-x^{1/3})^2 \rightarrow$ even.

PROPERTIES OF EVEN/ODD FUNCTION

① $f(x) = k$ is even function except $k=0$
② $f(x) = 0$ is even/odd function.
③

f	g	$f \pm g$	$f \times g$	f/g
Even	Even	Even	Even	Even.
O	O	O	E	E
E	O	NENO.	O	O

④ fog (ek baar bhi even dikhgya toh even)

$E(E) = \text{Even}$
$E(O) = \text{Even}$
$O(E) = \text{Even}$
$O(O) = \text{Odd}.$

$f = \sin x$ (odd) $\quad g = |x|$ (even)

$fog = \sin|x|$
$= \text{even}$

$fog(-x) = \sin|-x|$
$= \sin|x|$
$= fog(x)$

⑤ Any function containing origin in its domain can be written as sum of even & odd function both

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{odd}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{even}}$$

Ques. $f(x) = \log(x + \sqrt{x^2+1})$
$f(-x) = \log(-x + \sqrt{x^2+1})$
If NENO lage
check for

$f(x) + f(-x) = 0 \rightarrow$ Odd.
$\log(x+\sqrt{x^2+1})(-x+\sqrt{x^2+1}) = 0$
$\log(x^2+1-x^2) = 0 \rightarrow$ Odd

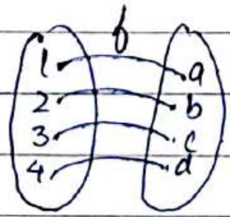
$f(x) - f(-x) = 0 \rightarrow$ Even

<u>CRUX</u> # $f(-x) = -f(x) \longrightarrow$ odd $(f(x))$
$f(x) + f(-x) = 0 \rightarrow$ odd
$f(x) - f(-x) = 0 \rightarrow$ even $f(x)$

$f(x)$
$\boxed{f(x) + f(-x)} \rightarrow$ Even
This whole function

$\boxed{f(x) - f(-x)} \rightarrow$ Odd
This whole function

INVERSE OF FUNCTION.
Let $f: A \rightarrow B$ be any one-one & onto (when whole co-domain is used)
function then its inverse $f^{-1}: A \rightarrow B$
is defined as
$$f^{-1}(y) = x \quad \text{iff} \quad f(x) = y$$



f^{-1} exist here but if
any element in co-domain
which left & does not
have any image in domain
for f^{-1}

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EQUALITY OF FUNCTION

$f(x)$ & $g(x)$ are said to be equal iff
$\downarrow$ D_f , $\downarrow$ D_g

(1) $D_f = D_g$
(2) $f(x) = g(x) \quad \forall \quad x \in D_f = D_g$.

Ques $f(x) = \dfrac{x^2+4}{x^2+4}$ & $g(x) = 1$

$D_f = R$ $\qquad D_g = R$

$\therefore$ Equal functions

Ques. $f(x) = \dfrac{x^2-1}{x^2-1}$ $\qquad g(x) = 1$

Domain $\qquad R$
$R - \{1, -1\}$

Domain not same
No Equal

Ques. $\sin^{-1}(\sin x)$ & $\sin(\sin^{-1} x) \rightarrow$ Not equal

$\sin x \rightarrow R$ $\qquad -1 \leq x \leq 1$

Ques $\log_e e^x$ $\qquad e^{\log_e x}$ $\qquad x^1 \longrightarrow$ Not equal

$e^x > 0, x \in R$ $\qquad x > 0$

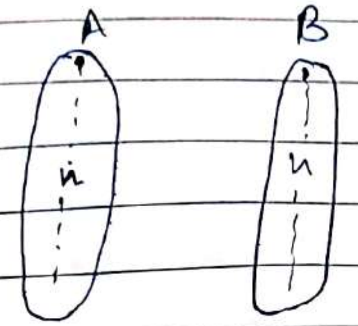
Ques. $\log_e x^2$, $2 \log_e x \longrightarrow$ not equal

$x^2 > 0$ $\qquad x > 0$
$x \in R - \{0\}$

No. of function from $A \to B$

$n(A) = m = |A| \qquad n(B) = n = |B|$

* Total no. of function $= n^m$
$= |B|^{A}$ ↳ cardinality



$n \times n \cdots n_{c,}$ ~~$m+n$~~ also have n options

n^n

* Total no. of constant function $= |B| = n(B)$
$= n$

cardinality of B.

no. of elements in B.

* Total no. of one-one function : $|A| \le |B| \to {}^nC_m \times m! \quad {}^nP_m$

$|A| > |B| \to 0$ (Not a one-one function)

$$:- \begin{cases} {}^{|B|}P_{|A|} & |A| \le |B| \\ 0 & |A| > |B| \end{cases}$$

* No. of onto function :- $|B| > |A| \to 0$

$|B| \le |A|$

$\sum_{r=1}^{|B|} (-1)^{|B|-r}\ {}^{|B|}C_r\ r^{|A|}$

$n(A)=a$ $\quad n(B)=b$

* No. of functions.

Codomain = Range

Case	Total	Onto	One-One
$n(A)=n(B)$	b^a	$a!$	$a!$
$n(A)>n(B)$	b^a	Grouping	0
$n(A)<n(B)$	b^a ☆	0	${}^bP_a = \frac{b!}{(b-a)!}$

CRUX ladke = ladki, b^a

One-One $a!$

Onto $a!$

Grouping

ladke → ladkian km (option km)

group banao. Onto

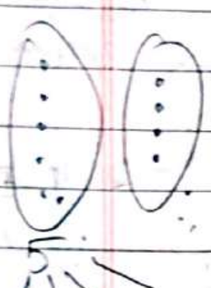
$n=2 \qquad 2^n-2$

$n=3$

$$\underbrace{3^n}_{\text{Total}} - \underbrace{{}^3C_1\times 2^n}_{\text{Exclusion}} + \underbrace{{}^3C_2\times 1^n}_{\text{Inclusion}}$$

$n=4$

$$\underbrace{4^n}_{\text{Total}} - \underbrace{{}^4C_1\times 3^n}_{\text{Exclusion}} + {}^4C_2\times 2^n - {}^4C_3\times 1^n$$



5

2 1 1 1

$$\left(\frac{5!}{2!\times1!\times1!\times1!\times3!}\right)4!$$

Not Onto:

Total $= b^a$

Onto = Grouping

Not = Total − Onto

Imp

$$f(x)+f\left(\frac{1}{x}\right) = f(x)\,f\left(\frac{1}{x}\right)$$

$$f(x) = 1 \pm x^n$$